Data Center Air Quality Analysis

Archived Data Report - Virginia DEQ, Office of Air Quality Monitoring

Loudoun County Study Area

October 22, 2026

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1 Introduction

As part of a special data center air monitoring project through a grant from EPA, Virginia DEQ's Office of Air Quality Monitoring (AQM) deployed APEX/Kunak AIR Pro sensors at nine locations in Loudoun County north of Dulles International Airport in the vicinity of Sterling and Ashburn.

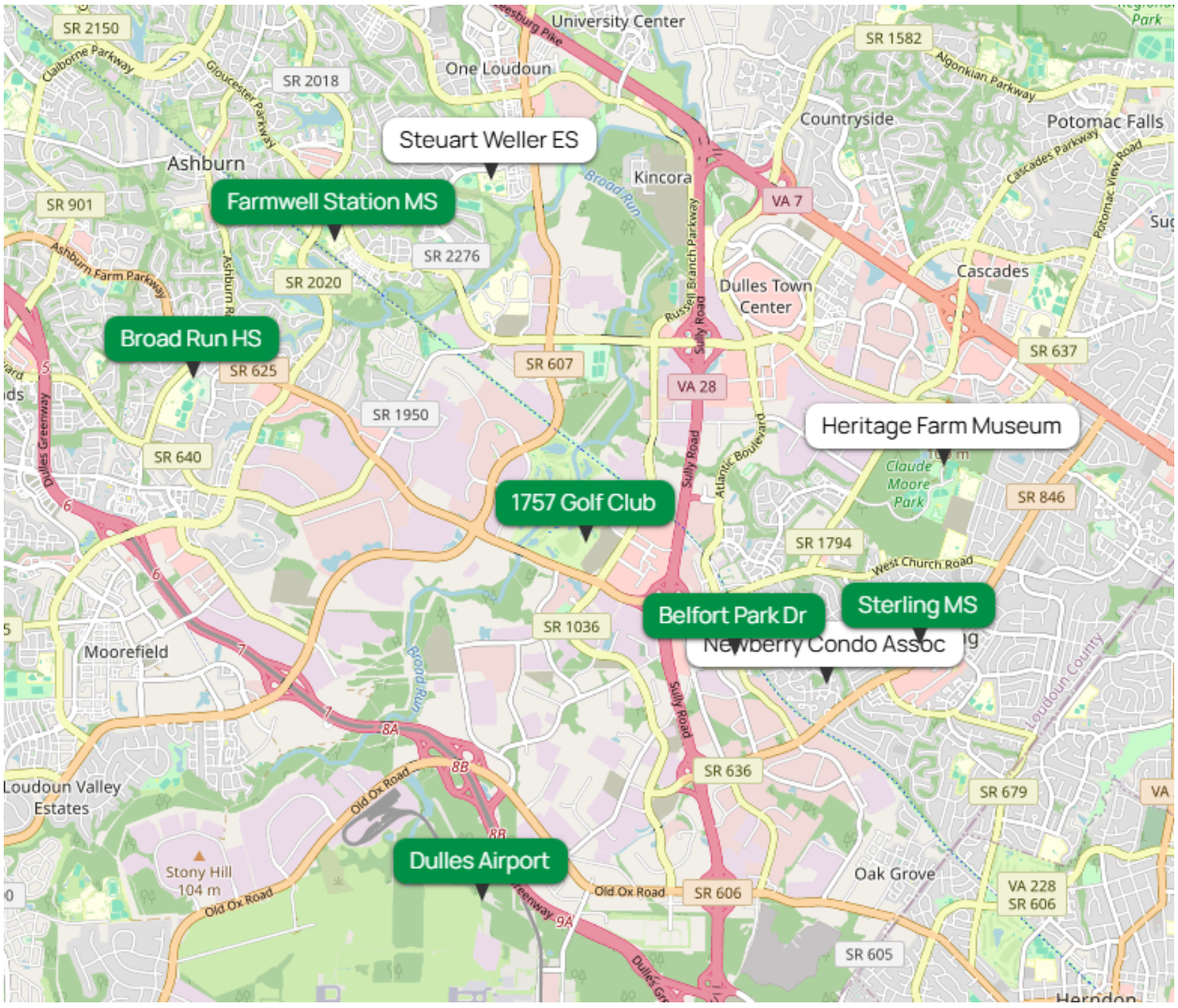


Figure 1: Study area with the APEX/Kunak sensors highlighted in green.

The pollutants that were measured are carbon monoxide (CO), nitrogen dioxide (NO_2), and particulate matter less than 2.5 microns in size ($PM2.5$).

These sensors are capable of collecting pollutant concentration data and establishing pollutant concentration trends, but they are not regulatory instruments as specified in the Code of Federal Regulations (CFR). They cannot be used in determining attainment or

non-attainment of the National Ambient Air Quality Standards (NAAQS). Only Federal Reference/Equivalent Method designated monitors can.

2 Sensor Timeline

- On Feb 25, 2026, Five sensors were installed in Loudoun County: Newberry Condo Association (Newberry Condo Assoc.), Broad Run High School (Broad Run HS), Dulles Airport, the Heritage Farm Museum, and Steuart Weller Elementary School (Steuart Weller ES). Due to the stabilization period, this report only uses data post-stabilization period, or from March 3, 2026 to September 17, 2026. [1]
- On April 8, 2026, two additional sensors were installed at 1757 Golf Course and Farmwell Station Middle School (Farmwell Station MS).
- On May 14, 2026, data collection at Newberry Condo Assoc. ended as the APEX/Kunak sensor 14 was moved to Belfort Park Drive (Belfort Park Dr.).
- On June 18, 2026, data collection at Heritage Farm Museum ended after the APEX/Kunak sensor 8 was removed for maintenance. Additionally, data collection at Steuart Weller ES ended as the APEX/Kunak sensor 6 was moved to Sterling Middle School (Sterling MS).
- On September 17, 2026, the data collection at 1757 Golf Course (sensor 13), Dulles Airport (sensor 9) Farmwell MS (sensor 10), and Sterling MS (sensor 6) ended as the sensors were moved from Loudoun County to Prince William County.
- On October 22, 2026, data collection at Dulles Airport (sensor 9) ended as this sensor was moved from Loudoun County to Prince William County. APEX/Kunak sensor 9 was moved to the Prince William Police Department West (PW Police Dept West). Additionally, APEX/Kunak sensor 5 at Broad Run HS was swapped out with APEX/Kunak sensor 8. APEX/Kunak sensor 5 was then installed at Meadows Farms Nursuries in Prince William
- *While there are sensors in Prince William and Loudoun County currently sampling, this archived report only contains data of Loudoun County sensors prior to October 23, 2026. When data collection for current Loudoun County sampling sensors ends, their full deployment data history will be added to this report.

Current ongoing data can be found in the weekly published Preliminary Data Analysis report.

[1] Kunak states that there is a stabilization period after installation, which may last up to 5 days. Only post-stabilization period data is analyzed in this report.

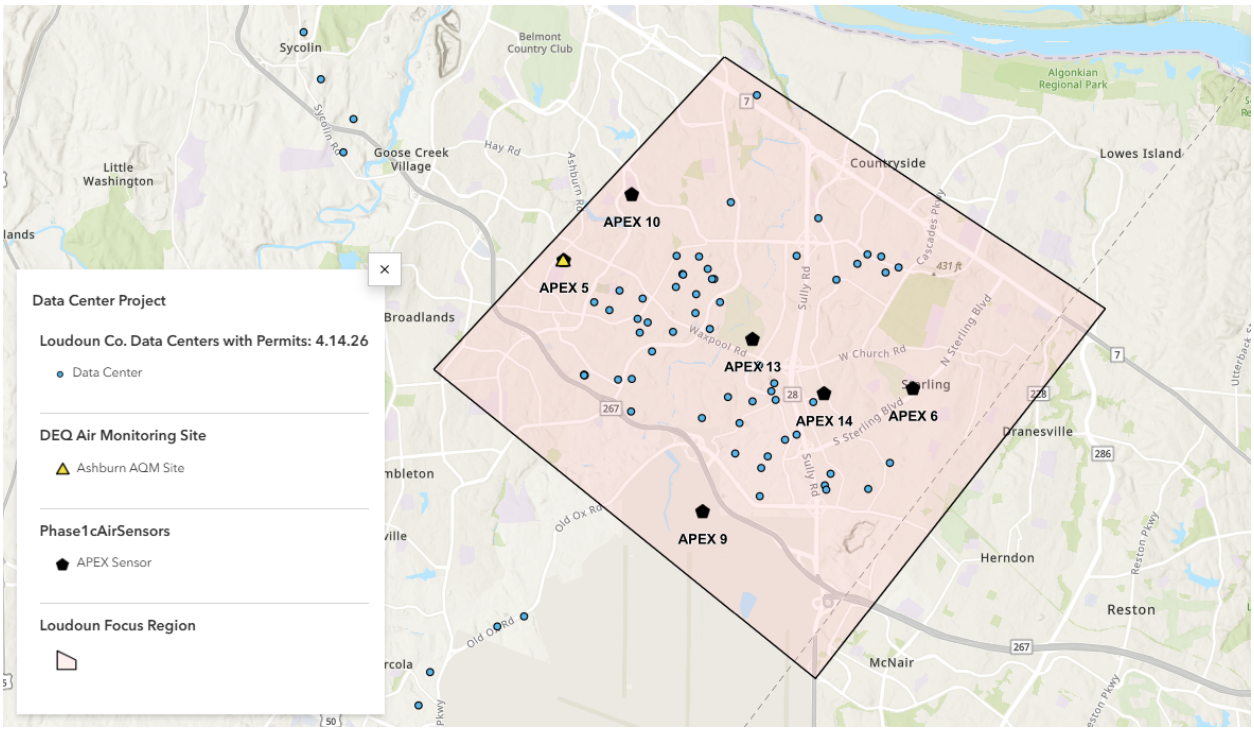


Figure 2: Study area highlighted, with permitted (blue circles) data centers mapped along with the local VADEQ air monitoring site (yellow triangle, Broad Run HS) and APEX/Kunak sensor positions (black pentagons).

The APEX Broad Run HS sensor is collocated (located at the same site) as Virginia DEQ's Broad Run HS regulatory monitoring site. The Broad Run HS regulatory monitoring site and the APEX/Kunak sensors both measure NO_2 and $PM2.5$, so a direct statistical comparison between these two instruments can be performed for these two pollutants.

For the APEX/Kunak CO sensors, the closest general ambient air regulatory monitoring site to the APEX/Kunak CO sensors is in Arlington County. Virginia DEQ's Arlington regulatory monitoring site is located outside of the "Data Center Alley" study area. There is a closer CO regulatory monitoring site in Fairfax County, but it is focused on near-highway and near-road emissions and would not be representative for general ambient air quality.

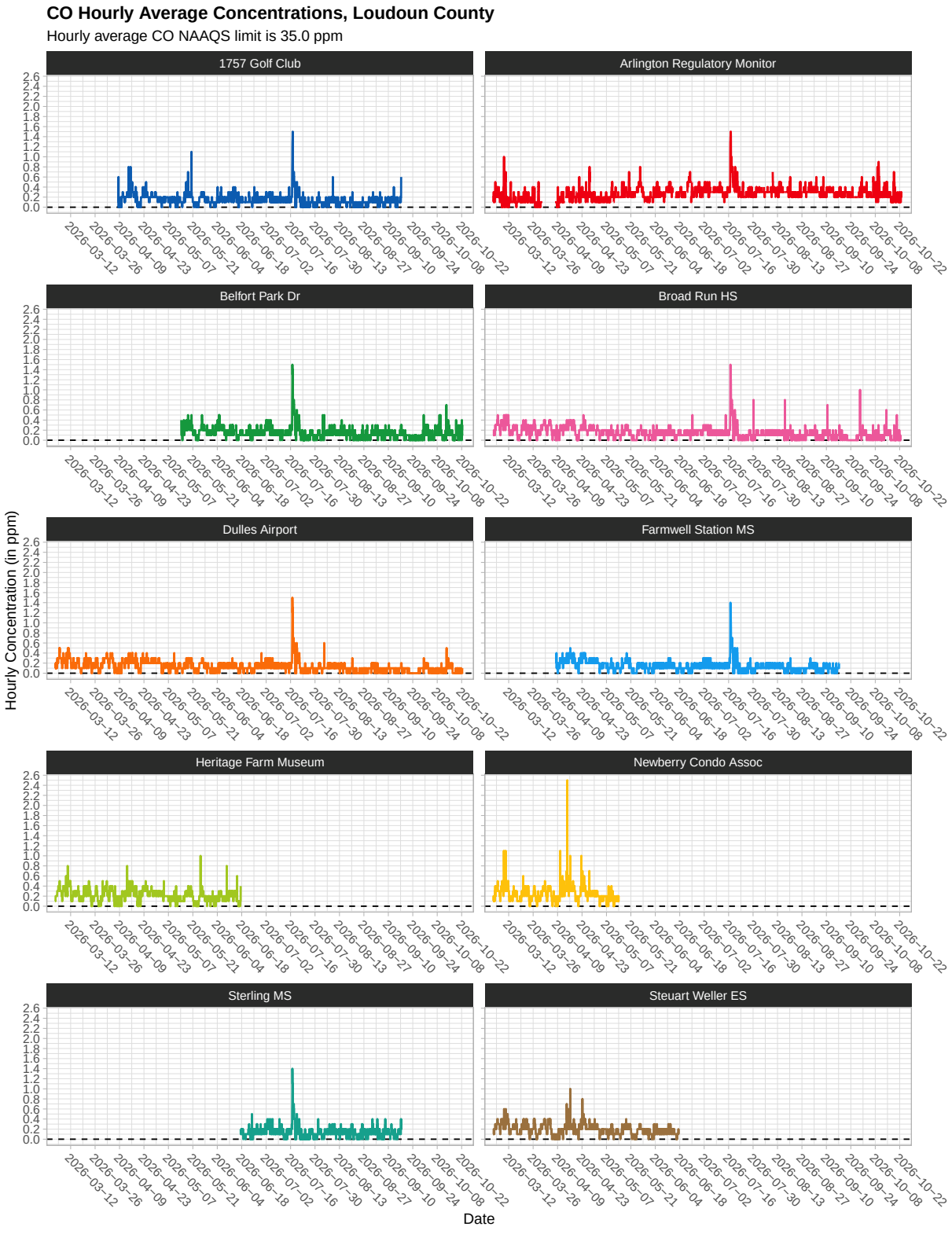
All dates & times presented below are in U.S. Eastern Standard Time (U.S. EST, UTC -5:00).

3 CO Hourly Concentrations: Loudoun County Sensor Data

The current National Ambient Air Quality Standards (NAAQS) for CO are set at two levels, corresponding to two different averaging times:

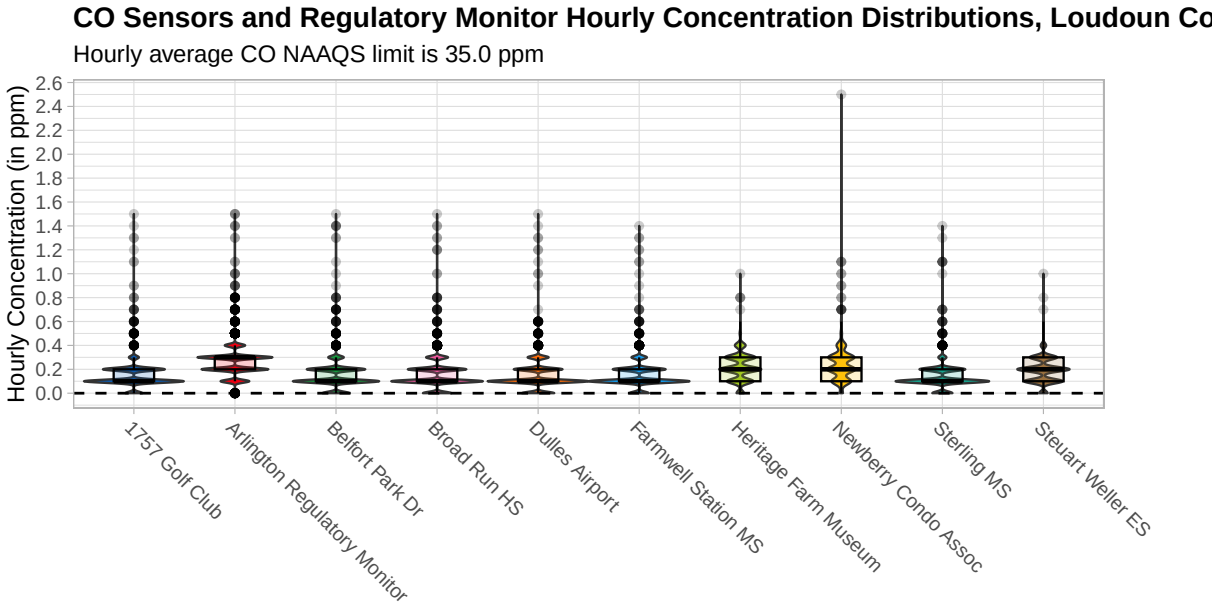
1. No 8-hour rolling concentration average may exceed 9 parts per million (ppm) more than once per year.
2. No 1-hour concentration average may exceed 35 parts per million (ppm) more than once per year.

The APEX/Kunak Air sensors measure CO in parts per billion (ppb), where 1000 ppb = 1 ppm. For ease of comparison, we have converted sensor ppb values into ppm to compare with our regulatory monitor values (which record in ppm, the unit that the CO NAAQS are assessed in). The closest comparable CO monitoring site to the APEX/Kunak sensors is Virginia DEQ's Aurora Hills monitoring site, in Arlington County, Virginia. According to the EPA, the largest sources of CO pollution are vehicles and other machinery that burn fossil fuels. As CO concentrations in Virginia have historically been well below the NAAQS levels (and very low in general), and because CO sources correlate with more traffic/higher population densities, the Virginia DEQ ambient air regulatory monitor for Northern Virginia is in Arlington County. This is why the Loudoun County regulatory air monitoring site (Broad Run HS) does not have a regulatory CO monitor.



As the graph above illustrates, CO concentrations from the sensors in Loudoun County are in the same general range as CO concentrations at the Arlington County regulatory

monitoring site, at around 0 - 1.1 ppm, with a few spikes up to 3.0 ppm.[2] The concentrations are well below the hourly NAAQS limit for CO (35.0 ppm). Because the hourly concentrations are consistently below 1 ppm, all 8-hour average concentrations are also well below the associated NAAQS limit of 9.0 ppm.The large spike near July 17, 2026, can be attributed to wildfire smoke from the wildfires occurring in Minnesota and Canada.



As the above violin plot (which plots a box plot over the statistical distribution of values across each site) illustrates, CO concentrations from the sensors in Loudoun County have relatively similar distributions and medians, in the 0.1 - 0.2 ppm range. A full table of statistical values from each sensor and the regulatory monitor can be found below. The 98th percentile values are extremely low across the region, all lying at or below 0.6 ppm.

[2] Due to data quality and instrumentation issues, data between March 31, 2026, and April 7, 2026, were voided at the Arlington regulatory CO monitor.

Table 1: CO Hourly Concentrations Statistical Summary, Loudoun County (in ppm)

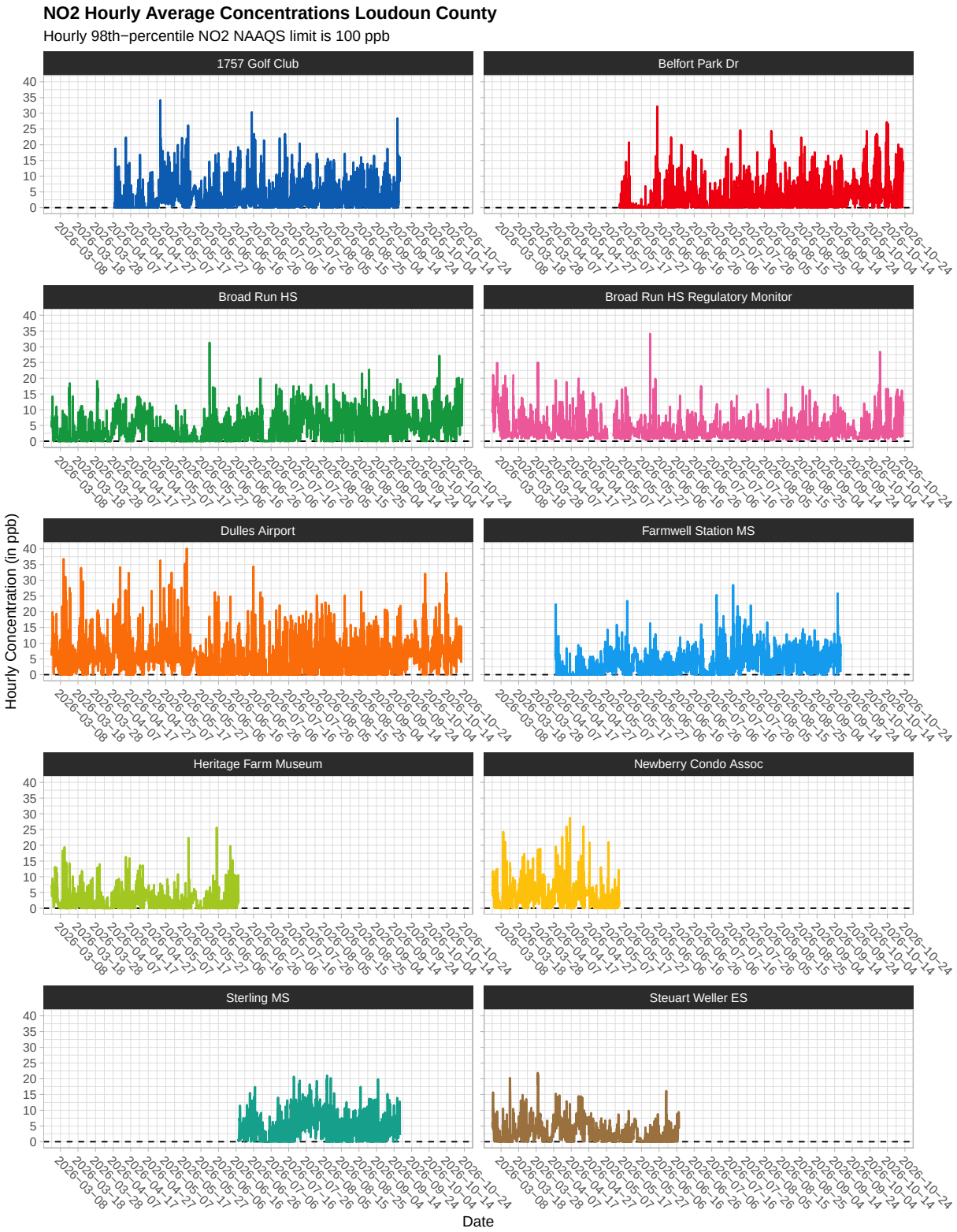
Site	mean	standard_deviation	median	percentile_98
1757 Golf Club	0.1	0.1	0.1	0.4
Arlington Regulatory Monitor	0.2	0.1	0.3	0.6
Belfort Park Dr	0.1	0.1	0.1	0.4
Broad Run HS	0.1	0.1	0.1	0.4
Dulles Airport	0.1	0.1	0.1	0.3
Farmwell Station MS	0.1	0.1	0.1	0.4
Heritage Farm Museum	0.2	0.1	0.2	0.4
Newberry Condo Assoc	0.2	0.1	0.2	0.5
Sterling MS	0.1	0.1	0.1	0.4
Steuart Weller ES	0.1	0.1	0.2	0.4

4 NO2 Hourly Concentrations: Loudoun County Sensor and Regulatory Monitor Data

The current National Ambient Air Quality Standards (NAAQS) for NO_2 are set at two levels, corresponding to two different averaging times:

1. The 98th percentile of 1-hour daily maximum concentrations of NO_2 may not exceed 100 parts per billion (ppb), when averaged over 3 years.
2. The yearly average of 1-hour NO_2 concentrations may not exceed 53 parts per billion (ppb).

The APEX/Kunak Air sensors measure NO_2 in parts per billion (ppb); Virginia DEQ's regulatory NO_2 analyzers also measure NO_2 in ppb. The closest NO_2 regulatory monitoring site is at Broad Run HS in Loudoun County, where a regulatory NO_2 monitor and an APEX/Kunak sensor (APEX Broad Run HS) are collocated.



The above time-series graph of shows a moderate amount of agreement between sensors, with the highest NO_2 values on average measured at the Dulles Airport NO_2 site. Due

to data quality and instrumentation issues, data between May 8, 2026, and May 10, 2026, were voided at the Ashburn regulatory NO_2 monitor. The full statistical analysis can be seen in the violin plot and the statistical summary table below:

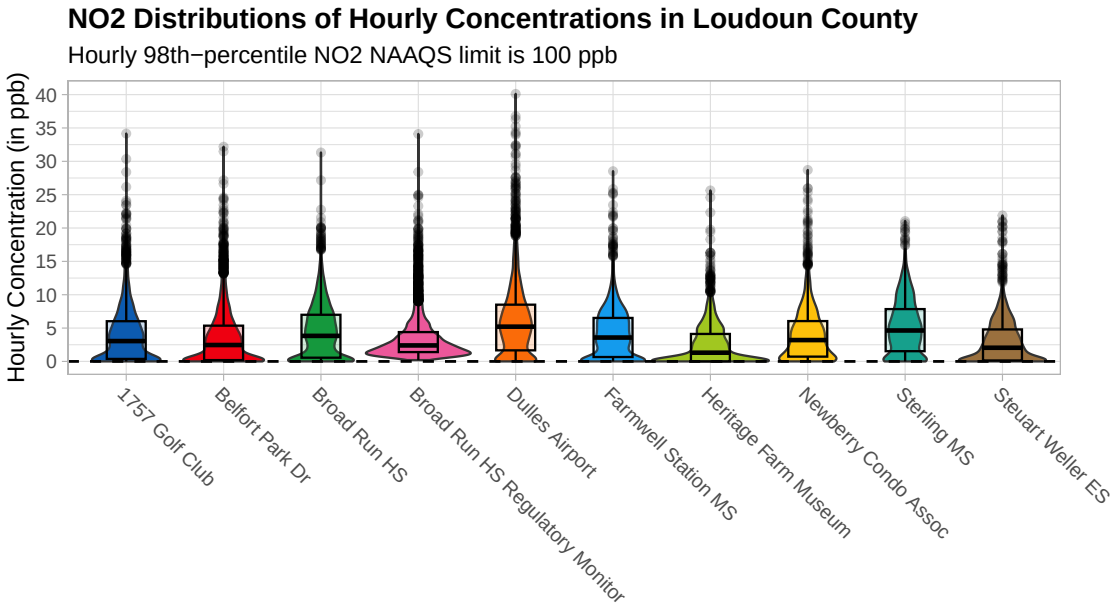


Table 2: NO2 Hourly Concentrations Statistical Summary, Loudoun County (in ppb)

Site	mean	standard_deviation	median	percentile_98
1757 Golf Club	3.9	4.1	3.0	15.3
Belfort Park Dr	3.6	4.2	2.5	15.7
Broad Run HS	4.4	4.0	3.8	14.1
Broad Run HS Regulatory Monitor	3.5	3.2	2.4	13.8
Dulles Airport	5.9	5.2	5.2	19.2
Farmwell Station MS	4.1	3.7	3.6	12.9
Heritage Farm Museum	2.5	3.1	1.3	10.8
Newberry Condo Assoc	4.1	4.1	3.2	15.8
Sterling MS	5.1	4.1	4.6	14.1
Steuart Weller ES	3.0	3.2	2.0	11.4

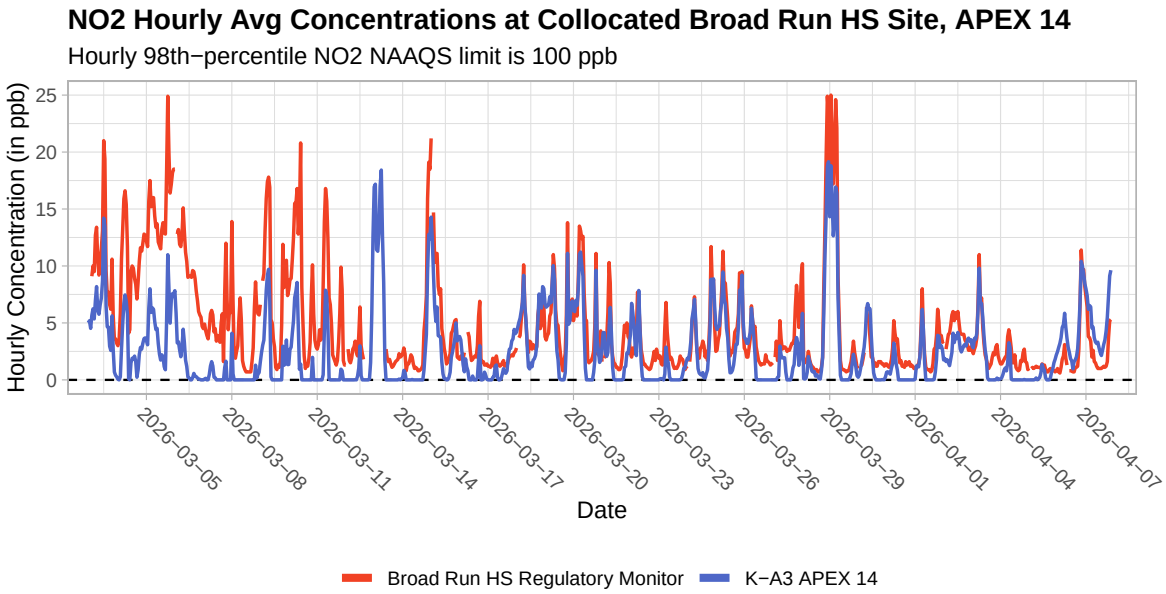
Mean values across all sites are below 10 ppb, with the highest at Dulles Airport and Sterling MS. The hourly 98th percentile values are low (in the 10 - 20 ppb range), compared with the NAAQS Hourly 98th percentile limit of 100 ppb.

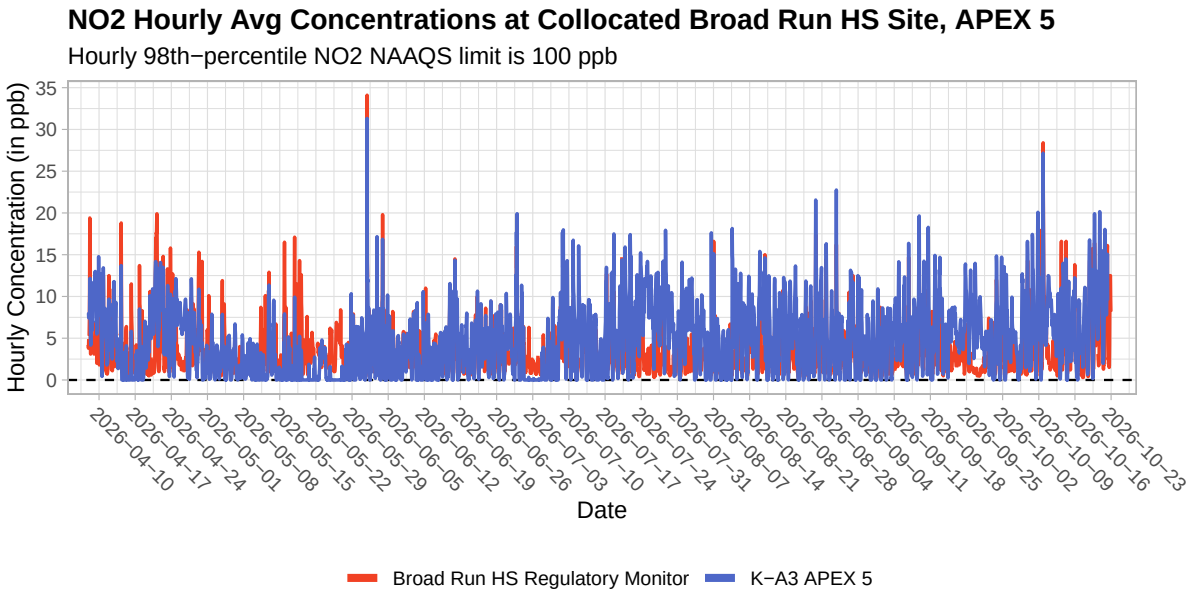
4.1 Collocation Statistics: Broad Run HS NO2

Because the Broad Run HS NO_2 regulatory monitor and the APEX/Kunak sensor at Broad Run HS are collocated, a deeper statistical analysis can be performed to compare the accuracy and correlation between the two analyzers. In order to get a sense of the general quality of the APEX/Kunak sensors, two sensors were rotated through to be collocated at Broad Run HS.

1. From March 3, 2026 - April 8, 2026, the APEX 14 sensor was collocated at Broad Run HS.
2. From April 8, 2026 - October 22, 2026, the APEX 5 sensor was collocated at Broad Run HS.

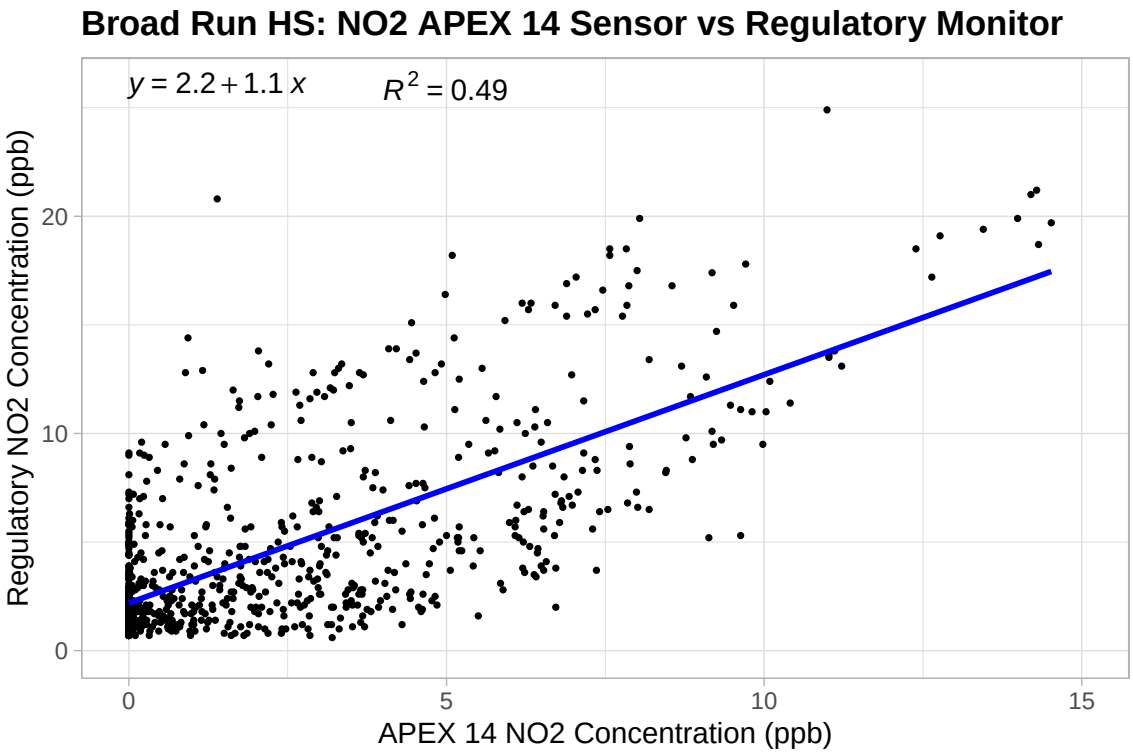
The data from each sensor was separated out and is presented below. First, the time-series graphs are compared to get a general view of agreement. In an ideal, perfect scenario the sensor data and the regulatory monitor data would lie completely on top of each other since they are measuring the same pollutant at the same place and time. Because no measurement system is truly perfect, and as the APEX/Kunak sensor relies on an entirely different NO_2 measurement method, there are some differences between the concentrations measured between the two analyzers, with the Broad Run HS regulatory monitor generally measuring higher concentrations compared to both sensors.

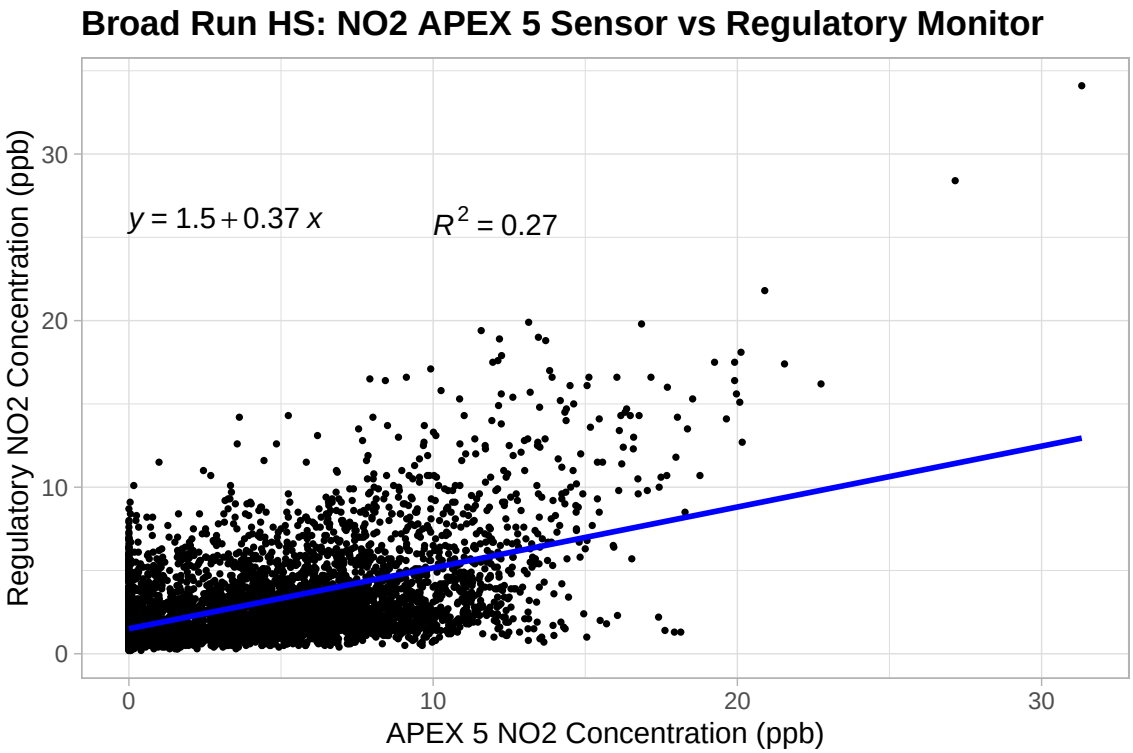




A scatter plot can also be used to compare concentrations measured by the APEX sensors with those from the Broad Run HS regulatory NO_2 monitor. In this plot, the APEX sensor readings are placed on the x axis and the regulatory monitor readings on the y axis. This allows for a visual assessment of how closely the two instruments agree. If the measurements matched perfectly, all points would fall along a single straight line.

The scatter plots below show that the APEX 5 sensor does not align closely with the Broad Run HS regulatory (T500u) monitor. This is reflected in the relatively low R^2 value, which indicates a weaker correlation and a poorer fit of the linear trend line. The best□fit line is shown in blue. In comparison, the APEX 14 sensor displays a stronger relationship with the regulatory monitor, as indicated by its higher R^2 value. APEX 5 has now had a longer time period correlated with the Broad Run HS regulatory monitor compared to APEX 14. Despite this time difference, APEX 5's R^2 value remains low. This could be attributed to difficulty in sensor accuracy when reading lower NO_2 concentrations.


Broad Run HS: NO2 APEX 14 Sensor vs Regulatory Monitor
y = 2.2 + 1.1 x
R2 = 0.49
Regulatory NO2 Concentration (ppb)
0
10
20
0
5
10
15
APEX 14 NO2 Concentration (ppb)


Broad Run HS: NO2 APEX 5 Sensor vs Regulatory Monitor
y = 1.5 + 0.37 x
R2 = 0.27
Regulatory NO2 Concentration (ppb)
0
10
20
30
0
10
20
30
APEX 5 NO2 Concentration (ppb)

5 PM2.5 Hourly Concentrations: Loudoun County Sensor and Regulatory Monitor Data

The current National Ambient Air Quality Standards (NAAQS) for $PM2.5$ are set at two levels, corresponding to two different averaging times:

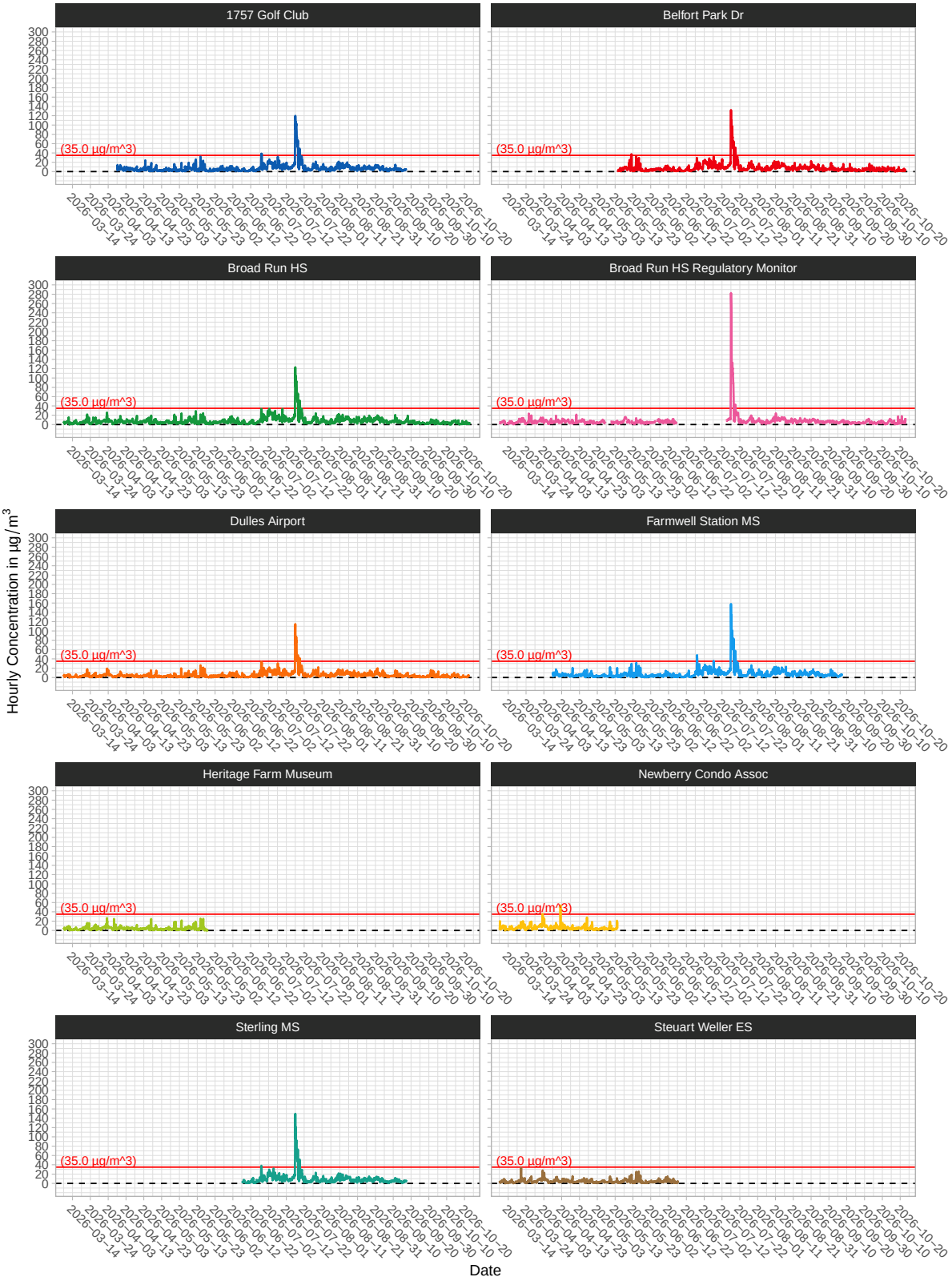
1. The yearly mean of 24-hr (daily) average $PM2.5$ concentrations may not exceed 9.0 micrograms per cubic meter ($\mu g/m^3$), when averaged over 3 years.
2. The 98th percentile of 24-hr (daily) average $PM2.5$ concentrations may not exceed 35.0 micrograms per cubic meter ($\mu g/m^3$), when averaged over 3 years.

The APEX/Kunak Air sensors and Virginia DEQ's regulatory $PM2.5$ monitors both measure PM2.5 in micrograms per cubic meter ($\mu g/m^3$). The closest $PM2.5$ regulatory monitoring site is at Broad Run HS in Loudoun County, which has a collocated regulatory $PM2.5$ monitor and an APEX/Kunak sensor.[3]

[3] Due to high humidity and data quality issues with the Kunak/APEX sensors, the first week of $PM2.5$ sensor collected data (March 3, 2026 - March 8, 2026) was voided and is not included in this analysis.

PM2.5 Hourly Average Concentrations in Loudoun County

24−hr Avg 98th−percentile PM2.5 NAAQS limit is 35.0 µg/m^3



Hourly concentrations greater than 35.0 $\mu g/m^3$ do not represent exceedances of the $PM2.5$ NAAQS, as NAAQS compliance is determined with the 98th percentiles of daily (24-hr) $PM2.5$ averages.

Besides some localized and short-term spikes, the $PM2.5$ data agrees very well between all the sensors and the Broad Run HS regulatory monitor. Due to data quality and instrumentation issues, data between May 7, 2026, and May 10, 2026, were voided at the Ashburn regulatory $PM2.5$ monitor. Additionally data at the Ashburn regulatory $PM2.5$ monitor between June 16, 2026, and July 14, 2026, were voided due to a failed data quality analysis test and instrumentation issues. $PM2.5$ data from the Heritage Farm sensor ended May 30, 2026, due to air flow failure in the particle sensor. $PM2.5$ levels were elevated surrounding the weekend of July 4, 2026. This is typical as fireworks greatly influence $PM2.5$ levels. There was a large spike through the weekend of July 17, 2026 due to the Minnesotan and Canadian wildfires. Further statistical analysis can be seen below in the violin chart and the summary statistics table:

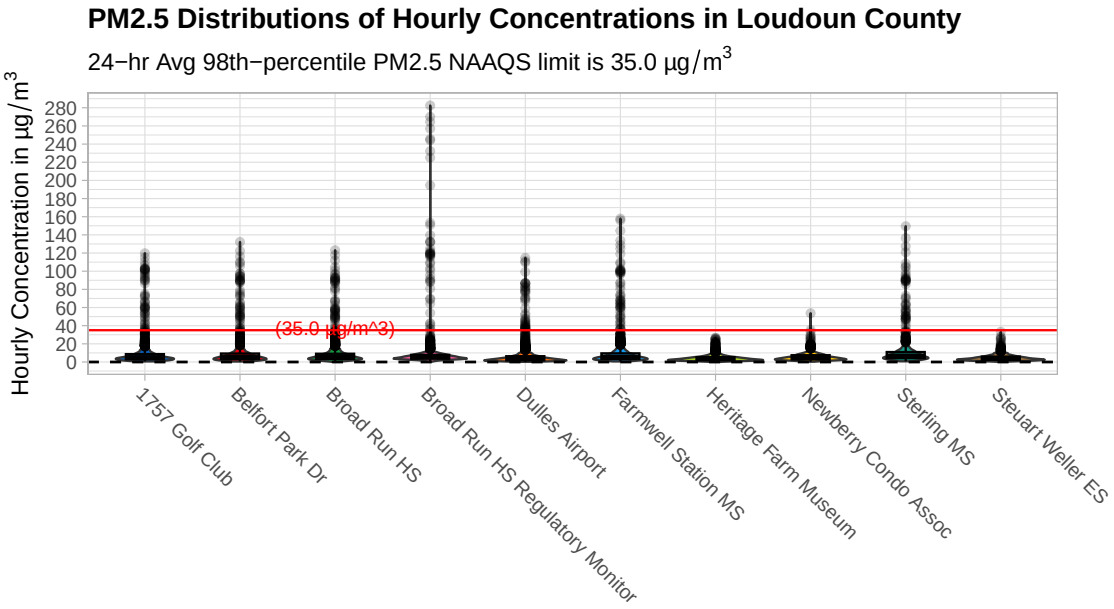


Table 3: PM2.5 Hourly Concentrations Statistical Summary

Site	mean	standard_deviation	median	percentile_98
1757 Golf Club	7.3	9.6	5.0	25.4
Belfort Park Dr	7.7	9.6	5.4	26.1
Broad Run HS	7.5	8.2	5.7	24.5
Broad Run HS Regulatory Monitor	7.4	14.2	5.4	19.4
Dulles Airport	5.2	7.0	3.5	18.4
Farmwell Station MS	8.0	10.7	5.5	26.4
Heritage Farm Museum	4.2	3.4	3.4	15.6
Newberry Condo Assoc	5.7	4.3	4.6	18.1
Sterling MS	9.4	11.9	6.8	47.1
Steuart Weller ES	4.5	3.2	3.7	13.5

The mean and median $PM2.5$ concentrations for each site are all below 10.0 $\mu g/m^3$. The Sterling MS sensor measured the highest mean concentration out of the group, followed by the Farmwell Station sensor. 98th percentile values are elevated due to the extreme $PM2.5$ values recorded during the Minnesotan and Canadian wildfires on the weekend of July 17, 2026. Although some Sterling's 98th percentile values appears to exceed the NAAQS 24-hr (daily) average $PM2.5$ concentration of 35.0 micrograms per cubic meter ($\mu g/m^3$), it does not as the NAAQS standard is based on a daily average across 3 years, not hourly averages.

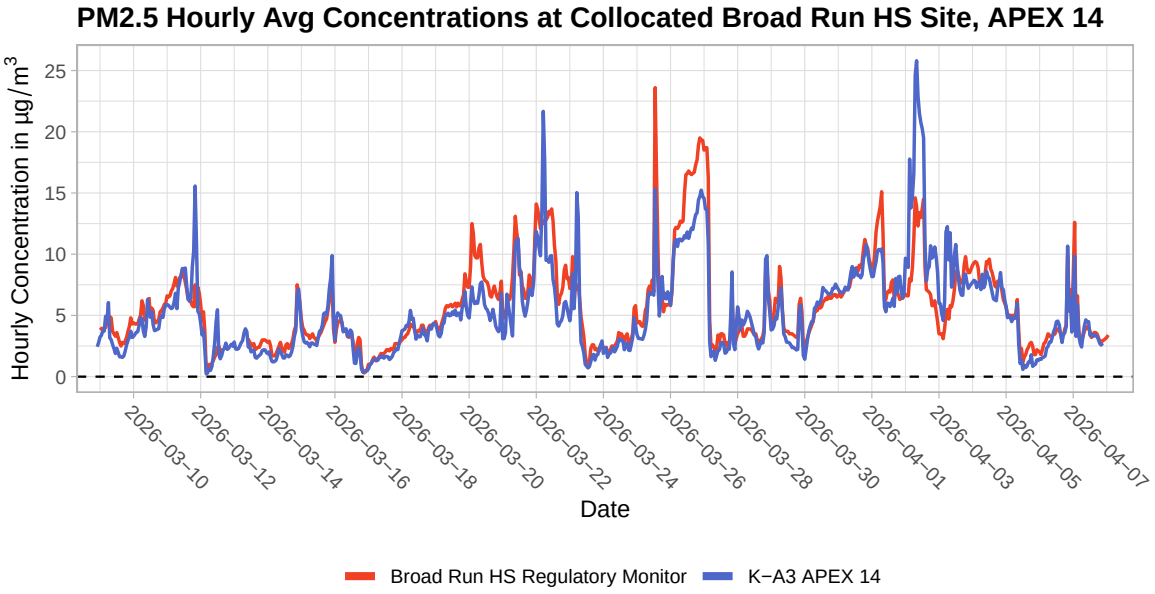
5.1 Collocation Statistics: Broad Run HS PM2.5

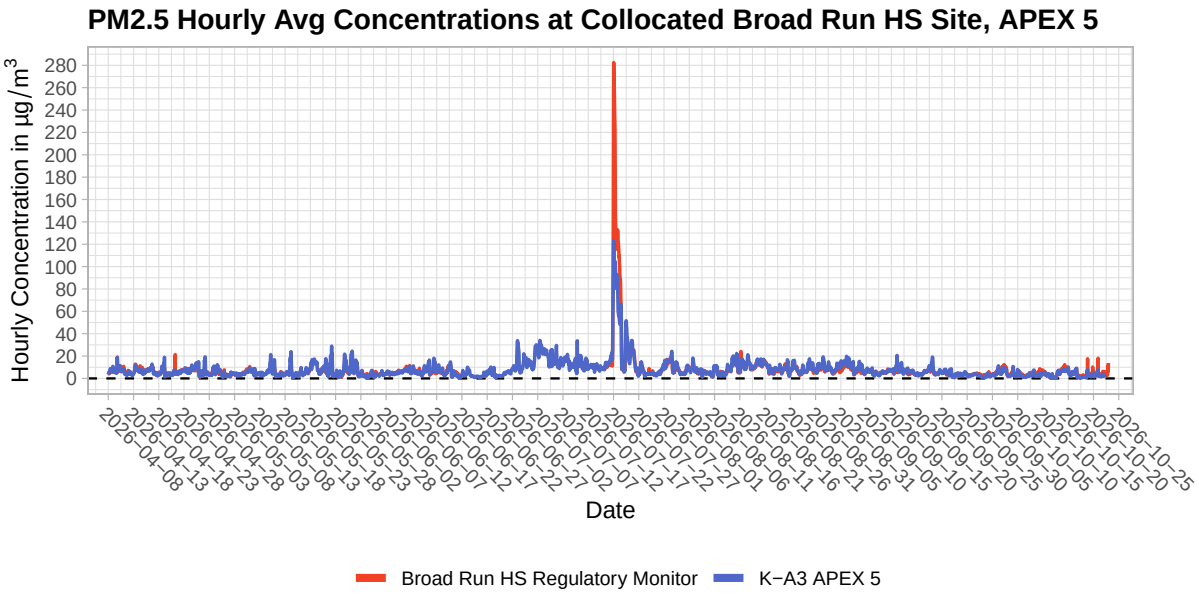
As with the NO_2 data, the APEX 14 sensor was collocated with the Broad Run HS $PM2.5$ regulatory monitor for a period of time before being swapped with the APEX 5 sensor - which is currently collocated with the Broad Run HS $PM2.5$ regulatory monitor.

The time-periods of collocation were the same for $PM2.5$ as they were for NO_2:

1. From March 3, 2026 - April 8, 2026, the APEX 14 sensor was collocated at Broad Run HS.
2. From April 8, 2026 - October 22, 2026, the APEX 5 sensor was collocated at Broad Run HS.

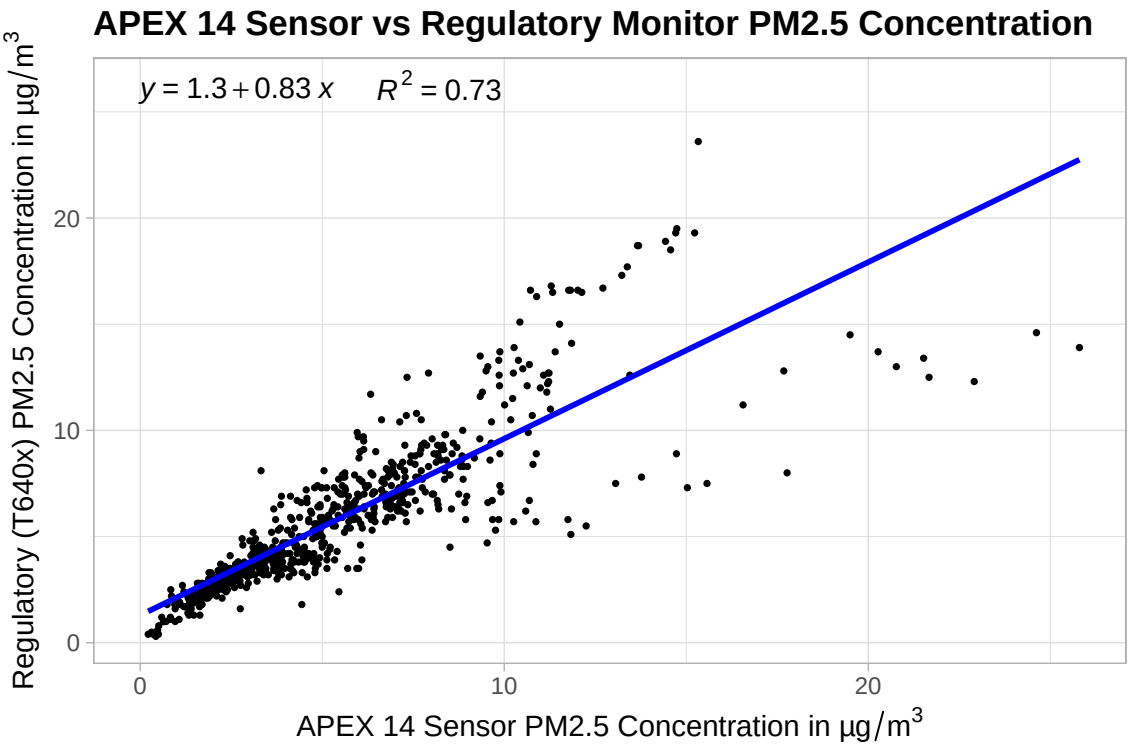
Again, we can compare the time-series to get a general view of agreement. In the graphs below, the readings from the APEX sensors and the regulatory $PM2.5$ monitor were overlaid on each other. As we can see in the graphs, the sensor data and the regulatory monitor are close, with the regulatory monitor reading slightly higher except for a few spikes in the sensor data.

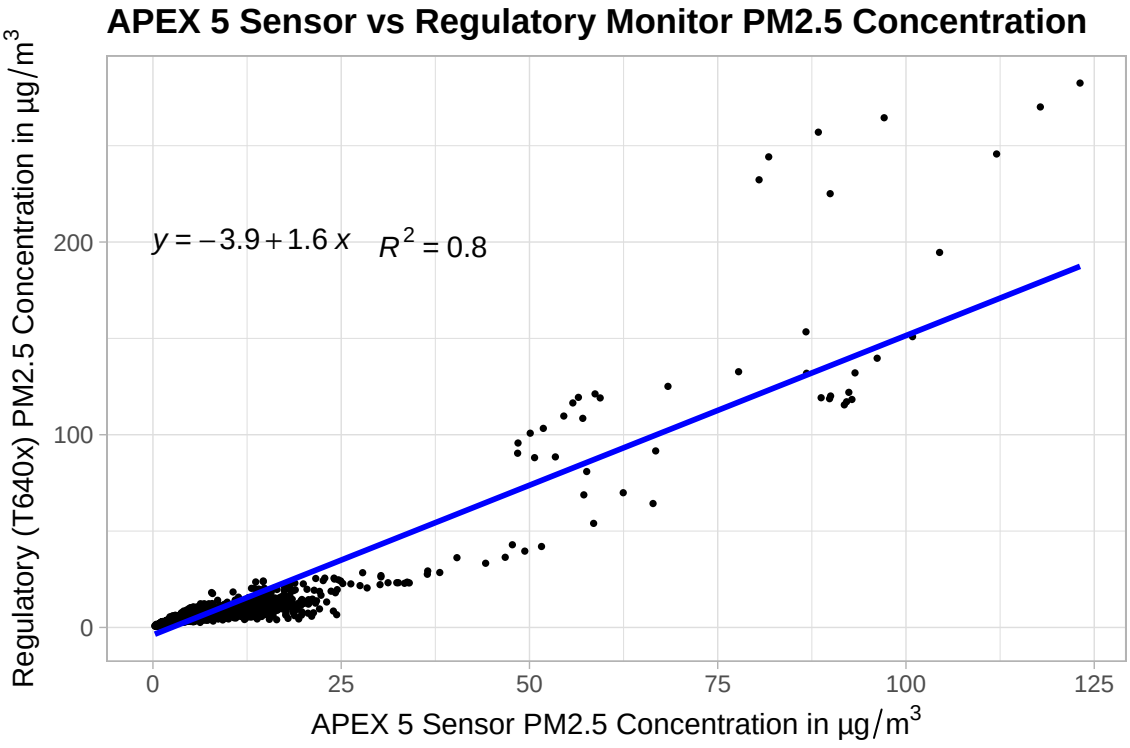




As with the NO_2 data, scatter plots were created for the APEX 5 and APEX 14 sensors. Both scatter plots show that data from the APEX $PM2.5$ sensors do not perfectly match the data from the Broad Run HS regulatory (T640x) monitor, although the $PM2.5$ data from the sensors are more accurate compared to the NO_2 sensor data.

Notably, the R^2 values for the PM2.5 data are much higher for both collocated APEX/Kunak sensors, indicating stronger correlations with the $PM2.5$ data between the APEX/Kunak sensors and the DEQ Broad Run HS regulatory monitor. At higher concentration values for $PM2.5$, differences between the sensor and regulatory monitor widen, as reflected in the scatter plots below. The recent wildfire impacted data from July 17, 2026, skew the linear relationship of APEX 5 to the regulatory monitor.


APEX 14 Sensor vs Regulatory Monitor PM2.5 Concentration
y = 1.3 + 0.83 x
R2 = 0.73
Regulatory (T640x) PM2.5 Concentration in µg/m3
APEX 14 Sensor PM2.5 Concentration in µg/m3
0
10
20


APEX 5 Sensor vs Regulatory Monitor PM2.5 Concentration
y = −3.9 + 1.6 x
R2 = 0.8
Regulatory (T640x) PM2.5 Concentration in µg/m3
APEX 5 Sensor PM2.5 Concentration in µg/m3
0
100
200
0
25
50
75
100
125

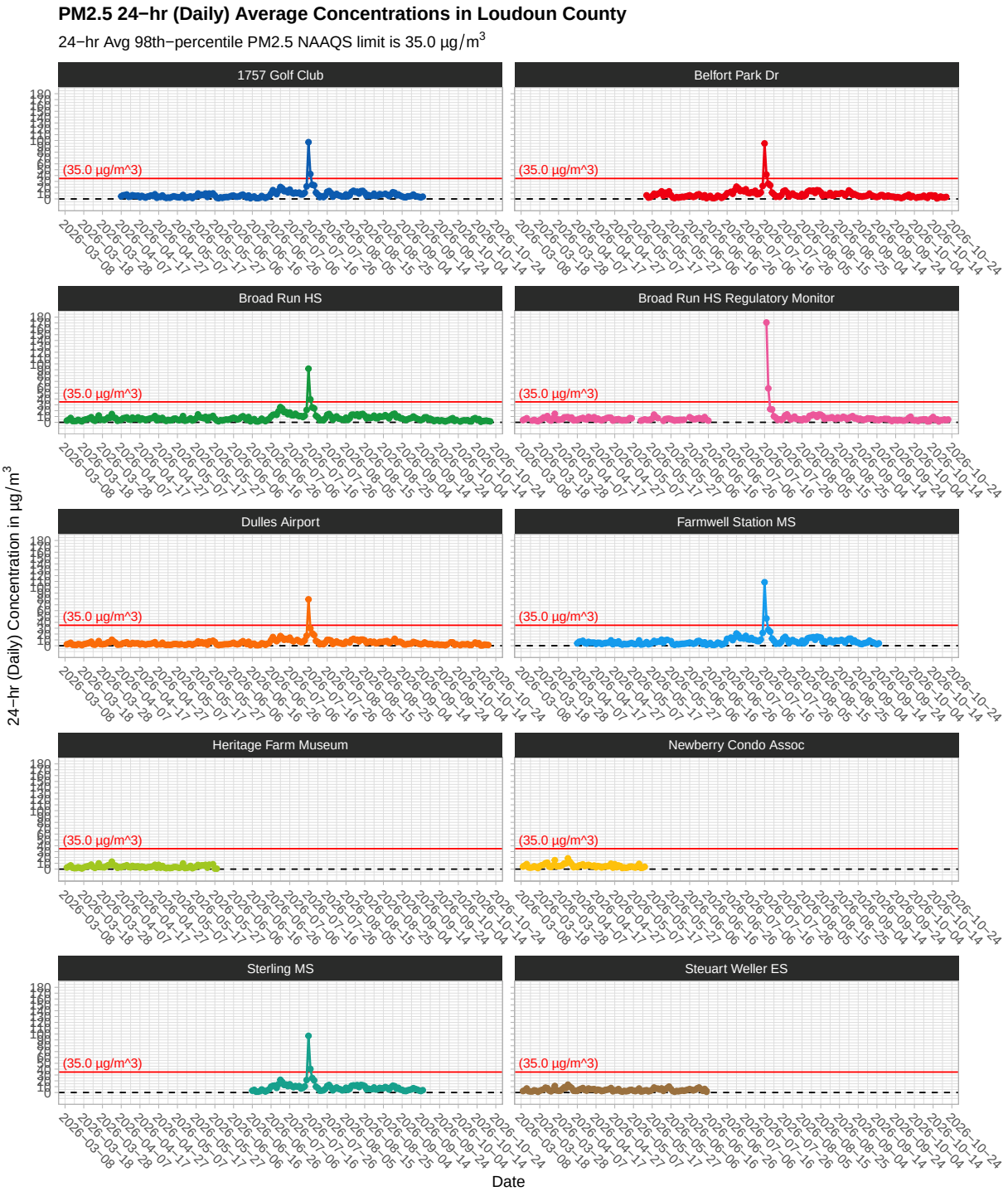
6 PM2.5 24-hour (daily) Concentrations: Loudoun County Sensor and Regulatory Monitor Data

As the $PM2.5$ NAAQS are given in terms of the 24-hour (daily) concentrations, the daily 24-hr average data is presented below, in a time-series and a distribution (violin plot) format.

The current National Ambient Air Quality Standards (NAAQS) for $PM2.5$ are set at two levels, corresponding to two different averaging times:

1. The yearly mean of 24-hr (daily) average $PM2.5$ concentrations may not exceed 9.0 micrograms per cubic meter ($\mu g/m^3$), when averaged over 3 years.
2. The 98th percentile of 24-hr (daily) average $PM2.5$ concentrations may not exceed 35.0 micrograms per cubic meter ($\mu g/m^3$), when averaged over 3 years.

At least 18 out of 24 hours are required to make a valid 24-hour (daily) $PM2.5$ average. This report contains data up to October 23, 2026.



Daily concentrations across all sensor sites and the Broad Run HS Regulatory monitor lie below the 35.0 $\mu g/m^3$ limit for $PM2.5$, with the exception of the wildfire impacted data near the weekend of July 17, 2026.

The violin plot below shows the distributions and median of all sensor data, as well as the

Broad Run HS regulatory monitor data. The full statistical analysis for the 24-hour (Daily) $PM2.5$ concentrations are presented in a table below the violin plot.

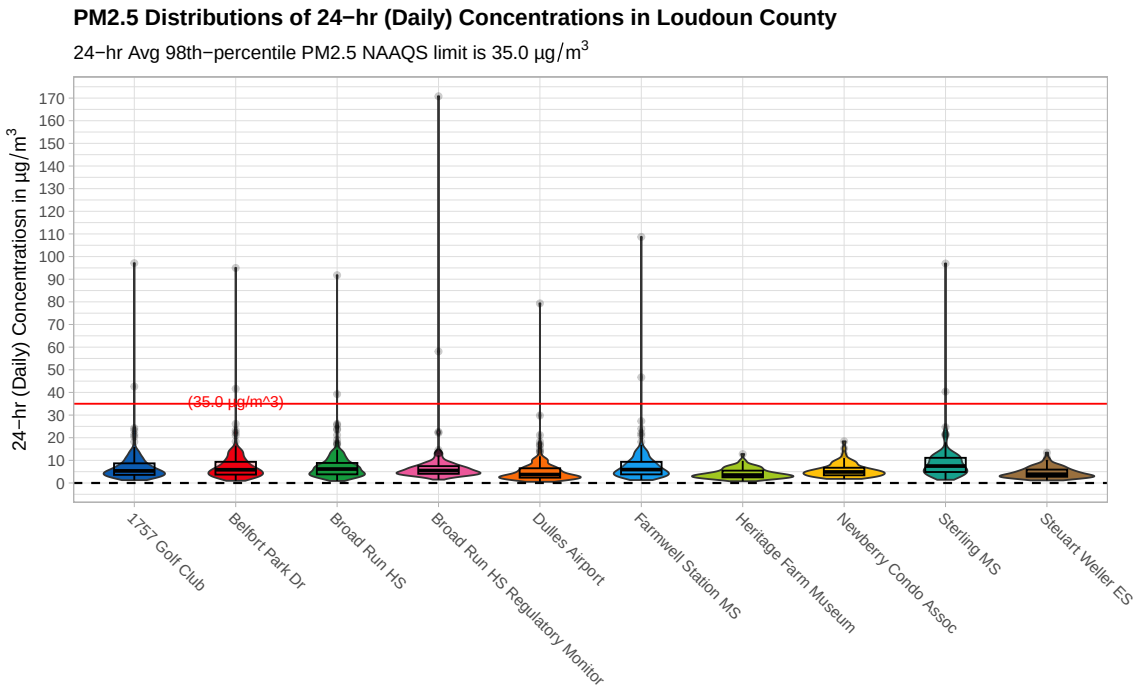


Table 4: PM2.5 24-Hour (Daily) Concentrations Statistical Summary

Site	mean	standard_deviation	median	percentile_98
1757 Golf Club	7.3	8.8	5.3	22.8
Belfort Park Dr	7.8	8.7	5.8	23.1
Broad Run HS	7.5	7.5	6.2	23.7
Broad Run HS Regulatory Monitor	7.3	13.0	5.5	16.7
Dulles Airport	5.2	6.3	3.7	16.8
Farmwell Station MS	7.9	9.7	5.9	23.6
Heritage Farm Museum	4.2	2.3	3.5	9.7
Newberry Condo Assoc	5.7	3.1	4.9	14.5
Sterling MS	9.4	10.8	7.5	27.5
Steuart Weller ES	4.6	2.4	3.9	10.2

As the above table illustrates, no site's 98th percentile concentration of the daily $PM2.5$ averages is above the 24-hr 98th percentile $PM2.5$ NAAQS limit of 35.0 $\mu g/m^3$. The Sterling MS sensor is above the 24-hr (daily) average $PM2.5$ NAAQS limit of 9.0 $\mu g/m^3$. Keep in mind that the NAAQS standards are based on 3 year averages, and so it cannot be said that these sensors are violating the NAAQS standards.

7 Wind Speed and Direction

Below is a wind rose from Dulles International Airport, directly South/Southwest of AQM's study area.

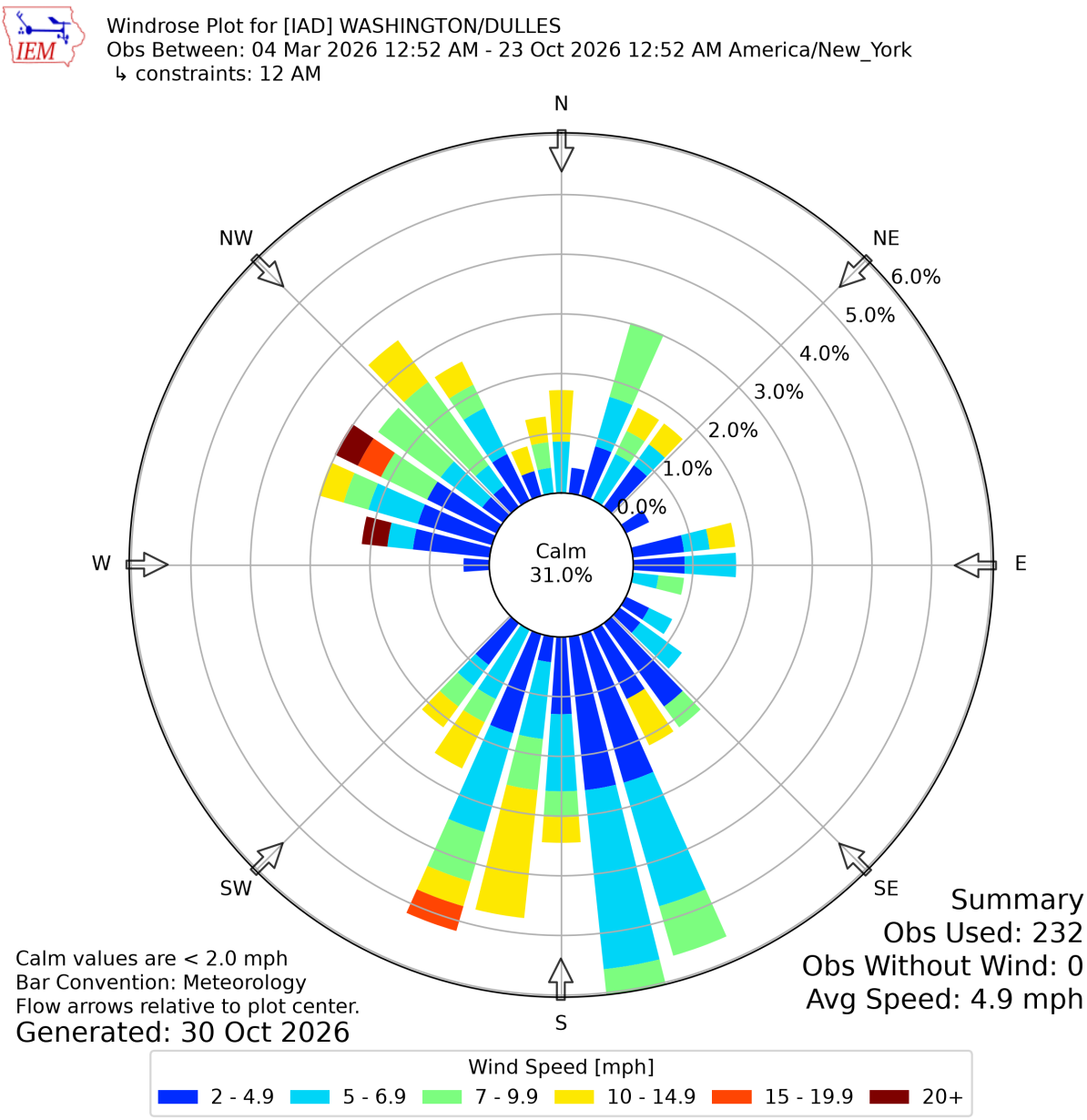


Figure 3: Wind Rose from Dulles Airport.

Wind speeds and directions during the study period show most winds came from either the South or Northwest direction. This suggests that AQM's Broad Run HS and Dulles sites are good upwind sites with regard to Loudoun County's "Data Center Alley".